Certification, Realization, and Obstruction:

A Universal Fiber Architecture

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Abstract

This paper closes the editorial loop for a multi-manuscript program in machine-checked mathematics. The executive claim is that a single *collapse / refinement / fiber* architecture—certification at a bare carrier, strict comparison upstairs, obstruction organized as fiber and section data—is instantiated internally in dependent type theory, validated on twelve external mathematical families, and then discharged into algebra, arithmetic (embedding problems), topology (Quillen for Galois connections), and logic/computability (self-certification and halting anchors). We state the role of each companion paper, a cross-domain dictionary, route labels used in the program (C/A/B/D), and what remains open. The intended reader is an advanced mathematician who has not read the series; citations point to the formal manuscripts and the Lean repository.

Keywords: formal verification, certification vs. realization, fibers, obstruction, Infinity Compression, Lean 4, program architecture

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1 Executive statement

One-sentence thesis. Canonical certification at a collapsed bare carrier does not exhaust structurally significant information: the gap is organized by a forgetful map, nontrivial fibers, and explicit obstructions—and that pattern repeats across internal formalization, external benchmarks, and classical mathematical domains.

2 Certification, realization, and non-exhaustion

Certification. A *certification* layer is a theorem-level invariant packaged as a canonical record (proof-irrelevant at that carrier): many routes may exist, but the official statement collapses to a unique bare certificate.

Realization. *Realization* refers to proof-relevant or structured data above that collapse: witnesses, roles, sections, or fiber points that are mathematically load-bearing once the comparison map is fixed.

Non-exhaustion. *Non-exhaustion* is the proved claim that certification does not determine realization: distinct legitimate data can share the same bare image, organized by π and fibers.

3 Flagship internal theorem

The Infinity Compression flagship paper [1] is the *internal origin theorem*: collapse of standard extraction, reflective split, strict refinement of the typed forgetful map, and partial fiber classification—all in the library's native carriers.

4 External validation (transferability)

The twelve-tranche survey [2] establishes *transferability*: the same schema applies to quotients, products, dependent sums, coproducts, subtypes, orbit and ideal quotients, pushouts, classifiers, localization, pullbacks, and a non-surjective control. Its job is not theorem-level leverage in classical fields; it rules out idiosyncrasy.

5 Algebraic and arithmetic engine

The group-extension paper [3] is the *algebraic and arithmetic engine*: section cocycles, splitting iff trivial cocycle, embedding problems as extension splitting, local-to-global obstruction, and the documented bridge toward Galois-theoretic towers.

6 Topological discharge

Quillen's Theorem A for Galois connections [4] is the *topology discharge*: nerves, contractible fibers over down-sets, and homotopy-equivalence data at the simplicial level for the Galois-connection case.

7 Logic and computability anchor

Route D in the program supplies a computability-facing discharge: self-certification phenomena aligned with standard undecidability infrastructure (e.g. halting and Rice scaffolding in the Mathlib `ComputablePred` layer), making the analogy to classical incompleteness precise without identifying the main theorem with Gödel's syntax. Exact statements live in the Lean `RouteD/` and `Summit/` modules; this synthesis paper does not restate every definition.

8 Cross-domain dictionary

The following table is a navigational index, not a formal commuting diagram.

Domain	**Carrier intuition**	**Obstruction / fiber role**
Internal IC	Bare summit certificate vs. enriched split	Non-injectivity of π_A; canonical fiber
External T1–T12	Baseline vs. structured bundle	Fibers, sections, right inverses
Algebra	Group extension	Section cocycle; splitting
Arithmetic bridge	Embedding problem	Solvable iff trivial cocycle
Topology	Galois connection $l \dashv u$	Contractible down-set fibers
Computability	Self-certification predicates	Undecidability anchors

9 Routes C, A, B, and D (program labels)

- **Route C (headline):** reflective non-exhaustion at the flagship carriers.
- **Route A (infrastructure):** supporting general lemmas (e.g. collision patterns, forgetful infrastructure) used by the headline.
- **Route B (applications):** algebra, topology, arithmetic bridges—the general-method papers.
- **Route D (logic/computability):** halting and Rice-style scaffolding aligned with computability theory rather than informal metaphor.

10 Why this is a reusable research instrument

The architecture is portable because it isolates *where* collapse occurs, *where* comparison must move upstairs, and *what* invariant obstructs lifting—independent of any single application domain. Lean provides a checkable artifact; the paper suite separates internal origin, transferability, domain discharges, and summit-level claims.

11 What remains open

Full fiber classification in the flagship setting; complete simplicial homotopy formalization behind Quillen; full H^1/H^2 classification in Mathlib; stronger unified quantification across all carriers in a single declaration; publication and Mathlib submission scheduling.

Acknowledgments

Repository: https://github.com/novaspivack/infinity-compression-lean.

References

[1] N. Spivack. *Canonical Certification Does Not Exhaust Reflective Structure: Reflective Split, Strict Refinement, and Fiber Structure in Infinity Compression.* Companion manuscript, 2026.

[2] N. Spivack. *External Validation of a Positive-Closure Proof Architecture: A Multi-Tranche Survey.* Companion manuscript, 2026.

[3] N. Spivack. *Fiber Architecture for Group Extensions in Lean 4: Cocycles, Splitting, and the Cohomological Bridge.* Companion manuscript, 2026.

[4] N. Spivack. *Quillen's Theorem A for Galois Connections: A Machine-Checked Formalization in Lean 4.* Companion manuscript, 2026.

[5] N. Spivack. *Reflective Non-Exhaustion: Certification, Realization, and the Summit Program.* Program manuscript, 2026.